Exploratory Data Analysis of Online Job Postings

Understanding Patterns Associated with Recruitment Fraud

Recruitment / Employment Industry

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Python Data Analysis Project

1. Industry Overview

Selected Industry

Recruitment / Employment

Online job platforms connect employers and job seekers through job advertisements.

Industry Challenge

Job platforms can contain both genuine employment opportunities and fraudulent postings.

Role of Data Analysis

Data analysis helps examine job-posting characteristics and identify patterns associated with recorded fraud labels.

Project focus

Job characteristics and employment information

Company and recruitment-related indicators

Industry patterns

Text characteristics of job descriptions, company profiles and requirements

Statistical association between company-logo presence and the fraud label

2. Problem Statement

Online job platforms contain genuine employment opportunities as well as fraudulent postings.

Identifying patterns associated with suspicious job advertisements can help understand potential recruitment risks.

The project examines job-posting characteristics in relation to the recorded fraudulent label.

The analysis covers company profile availability, salary disclosure, employment type, job-description length, industry, location and recruitment-related indicators.

The project distinguishes observed patterns and statistical associations from evidence that a particular posting is fraudulent.

Affected stakeholders Job seekers • Recruitment platforms • Employers

3. Objectives and Proposed Solution

A reproducible EDA workflow

1. Prepare

Load, inspect, clean, and transform data.

2. Explore

Compare labels and selected features.

3. Communicate

Visualize patterns and explain limitations.

Proposed solution: Use Python-based data cleaning, exploratory analysis, visualization and statistical testing to examine patterns associated with the recorded fraudulent label.

4. Tools and Technologies

Tools used in the analysis

- Python — main programming language.
- Jupyter Notebook — interactive code, analysis, and outputs.
- pandas / NumPy — data loading, cleaning, transformation, and calculations.
- Matplotlib / Seaborn — visualizations
- SciPy / chi-square test — association test
- GitHub — repository hosting and project sharing.

5. Dataset Details

Describe the data before cleaning

- Initial combined data: 35,479 rows × 18 columns (reported in the analysis).
- Target variable: `fraudulent`; 0 represents genuine and 1 represents fraudulent.
- features: company profile, description, requirements, industry, employment type, has_company_logo, has_questions, and telecommuting.
- Files discussed in the project include `fake_job_postings.csv` and `Job_Fraudlent_Dataset.csv.

6. Errors and Issues in Raw Data

Data quality issues

- Missing values in several text and categorical columns.
- Duplicate records that could distort counts if not checked.
- Missing or invalid values in the `fraudulent` target column.
- Different levels of completeness across company profiles, requirements, and descriptions.
- Class imbalance: fraud-labelled records are much fewer than genuine-labelled records.

7. Data Cleaning and Transformation

- Load the CSV files and inspect shape, columns, data types, and sample rows.
- Measure missing values and check duplicate records.
- Clean relevant fields and handle missing values based on analysis needs.
- Use records with valid fraud labels for labelled comparisons.
- Create or inspect measures such as description word count and company-profile length.
- Recheck row counts and target-label counts after cleaning.

8. Data Analysis and Visualization

Label distribution

Compared genuine and fraudulent counts/percentages.

Categorical features

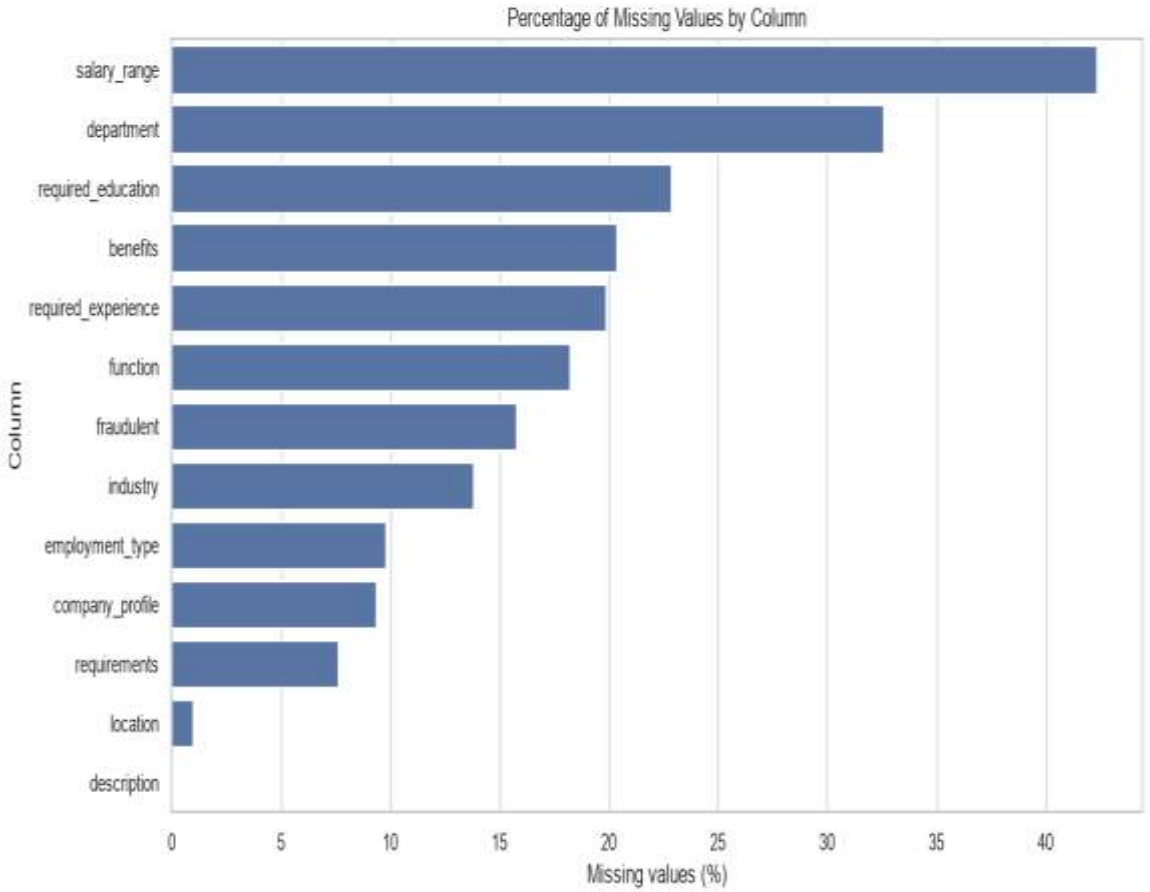
Compared fraud percentage by employment type, industry, logo, screening questions, and telecommuting.

Text completeness

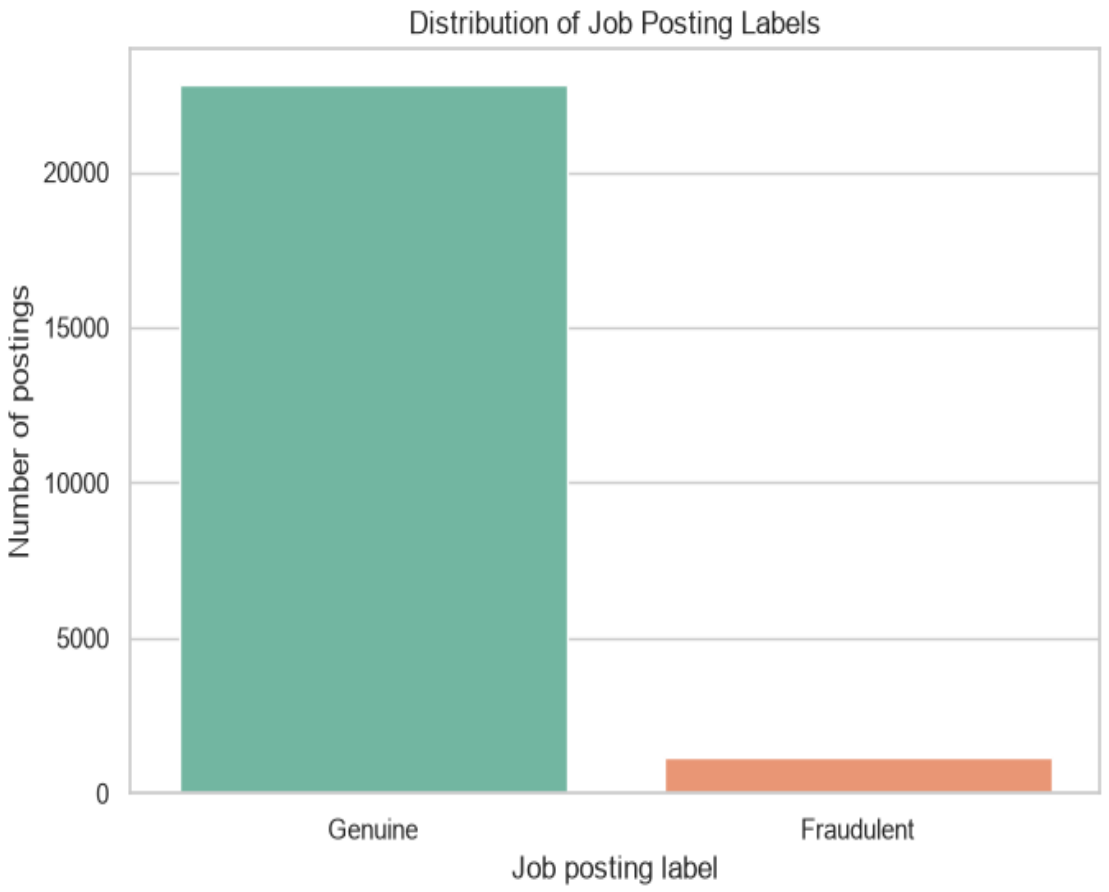
Compared description, company-profile, and requirements length.

Data Visualization – Actual Notebook Charts

The following charts are the visualizations embedded in the executed notebook.

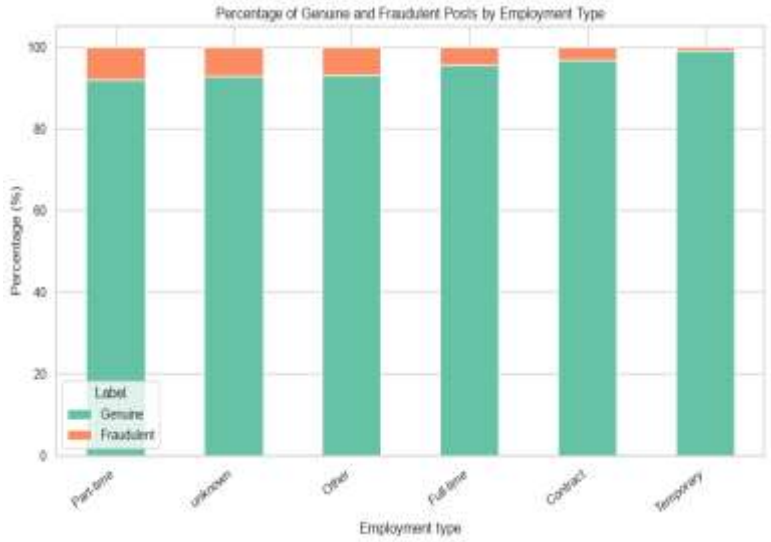


Missing values before cleaning → identifies columns requiring preprocessing.

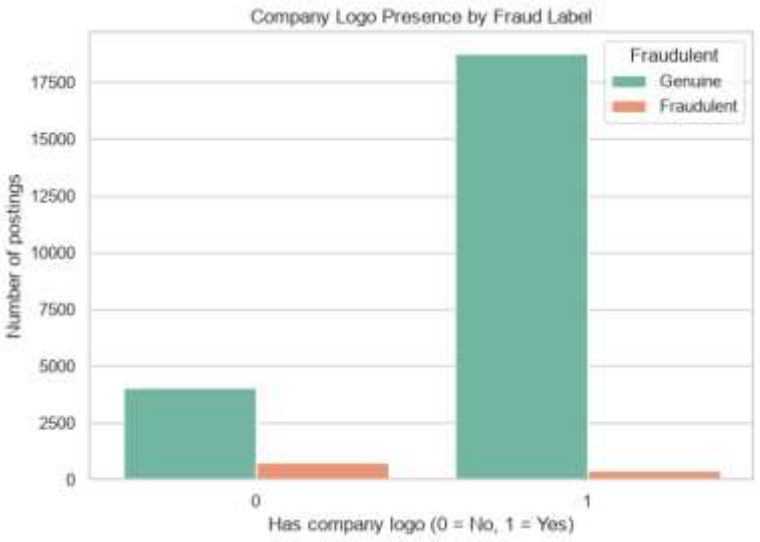


Genuine vs fraudulent postings → shows the recorded label distribution.

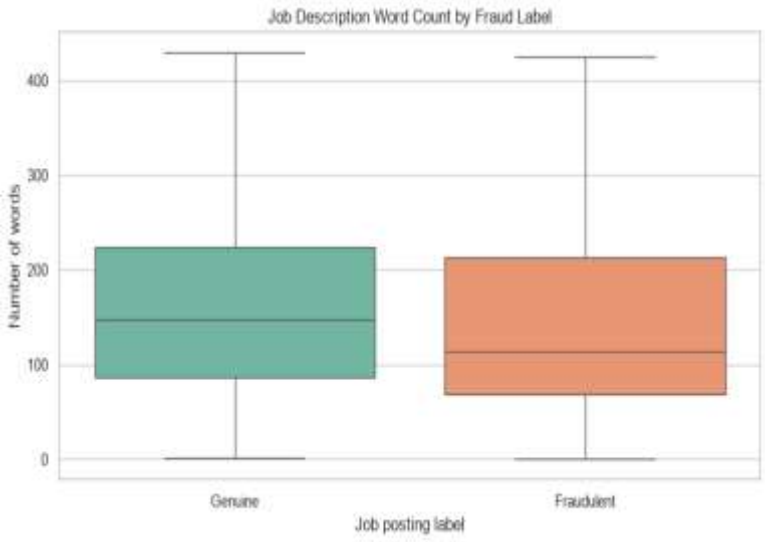
Data Visualization – Recruitment & Text Patterns



Fraud percentage by employment type → compares fraud proportions across employment categories.

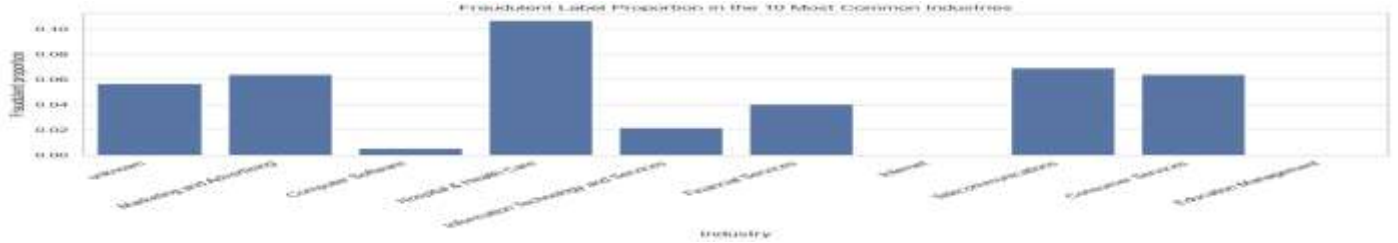


Company logo vs fraud label → compares the recorded fraud label by logo presence.



Job description length → compares description word counts by fraud label.

Industry visualization → shows fraudulent-label proportion across the 10 most common industries.



9. Key Insights and Findings

- Reported labelled subset: 22,805 genuine and 1,173 fraudulent records (95.11% / 4.89% of labelled records).
- Median description length: 147 words for genuine postings vs. 114 for fraudulent postings.
- Median company-profile length: 595 characters for genuine postings vs. 7 for fraudulent postings.
- Median requirements length: 483 characters for genuine postings vs. 249 for fraudulent postings.
- Logo presence was associated with the fraud label: $\chi^2 = 1592.44$, df = 1. Report $p < 0.001$ rather than $p = 0.0$ if the output was rounded.

10. Limitations and Responsible Interpretation

What the results can and cannot say

- EDA identifies associations; it does not prove causation.
- Results reflect this dataset and may not generalize to every job portal or current scam tactic.
- Missing labels and class imbalance affect interpretation.
- The project does not train or deploy a predictive model; that could be a future extension.
- Document the source, cleaning rules, and final row counts so the analysis can be reproduced.

11. GitHub Repository and Structure

Repository

github.com/priyadharshan-sivakumar/job-fraud-eda

Explain the project files

- README.md — overview, source, instructions, findings
- Dataset/ — raw and cleaned CSVs
- Python/ — loading, cleaning, analysis, visualization scripts
- Notebook — analysis cells and outputs
- Visualizations/ — exported charts, if present

Conclusion

EDA helps reveal differences between genuine and fraudulent job postings and highlights data-quality issues before any predictive model is built.

Thank you.

I’m happy to answer your questions.